

Maxwell's Equations in Vacuum: Derivation of the Wave Equations and the Poynting Vector

1. Starting point

In vacuum there are no sources: $\rho = 0$ and $\mathbf{J} = 0$. Maxwell's equations in differential form reduce to

$$\nabla \cdot \mathbf{E} = 0$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

or, using $c^2 = 1/(\mu_0 \varepsilon_0)$,

$$\nabla \times \mathbf{B} = \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t}$$

The four equations are, respectively:

- **Gauss (electric):** no free charge, so $\mathbf{E}$ is divergence-free.
- **Gauss (magnetic):** no magnetic monopoles, so $\mathbf{B}$ is divergence-free (always true).
- **Faraday's law:** a time-varying $\mathbf{B}$ induces a circulating $\mathbf{E}$.
- **Ampère-Maxwell's law:** in vacuum the displacement current $\mu_0 \varepsilon_0 \partial_t \mathbf{E}$ is the only source of a circulating $\mathbf{B}$ (there is no conduction current).

Two vector identities are used in what follows:

$$\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}$$

$$\nabla \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{B})$$

2. Wave equation for $\mathbf{E}$

Step 1 — Take the curl of Faraday's law. The curl and the time derivative commute (the fields are assumed smooth):

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t}(\nabla \times \mathbf{B})$$

Step 2 — Simplify the left-hand side with the curl-curl identity. The divergence term vanishes by Gauss's law in vacuum ($\nabla \cdot \mathbf{E} = 0$):

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E}$$

Step 3 — Substitute Ampère-Maxwell's law on the right-hand side:

$$-\frac{\partial}{\partial t}(\nabla \times \mathbf{B}) = -\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

Step 4 — Equate and rearrange:

$$\boxed{\nabla^2 \mathbf{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2}}$$

with $c^2 = 1/(\mu_0 \varepsilon_0)$. This is the wave equation written componentwise: each of E_x, E_y, E_z satisfies the same scalar wave equation.

3. Wave equation for $\mathbf{B}$

The argument is the mirror image. **Take the curl of Ampère-Maxwell's law:**

$$\nabla \times (\nabla \times \mathbf{B}) = \mu_0 \varepsilon_0 \frac{\partial}{\partial t}(\nabla \times \mathbf{E})$$

Left-hand side: $-\nabla^2 \mathbf{B}$, by the curl-curl identity and Gauss's law for magnetism ($\nabla \cdot \mathbf{B} = 0$).

Right-hand side: use Faraday's law, $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$:

$$\mu_0 \varepsilon_0 \frac{\partial}{\partial t} \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}$$

Therefore:

$$\boxed{\nabla^2 \mathbf{B} = \frac{1}{c^2} \frac{\partial^2 \mathbf{B}}{\partial t^2}}$$

4. Consequences: plane waves

Insert a monochromatic ansatz

$$\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$$

into the wave equation: $\nabla^2 \mathbf{E} \rightarrow -k^2 \mathbf{E}$ and $\partial_t^2 \mathbf{E} \rightarrow -\omega^2 \mathbf{E}$, giving the dispersion relation

$$\omega = c |\mathbf{k}|$$

Faraday's law then gives

$$\mathbf{B}_0 = \frac{\hat{\mathbf{k}} \times \mathbf{E}_0}{c}$$

and $\nabla \cdot \mathbf{E} = 0$ gives $\mathbf{k} \cdot \mathbf{E}_0 = 0$. Hence the waves are **transverse**: they propagate at speed c , with $\mathbf{E} \perp \mathbf{B} \perp \hat{\mathbf{k}}$ and $|\mathbf{B}_0| = |\mathbf{E}_0|/c$.

5. The Poynting vector

Step 1 — Energy density. The vacuum electromagnetic energy density is the sum of the electric and magnetic contributions:

$$u = \frac{1}{2} \varepsilon_0 E^2 + \frac{1}{2\mu_0} B^2$$

Step 2 — Differentiate in time:

$$\frac{\partial u}{\partial t} = \varepsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} + \frac{1}{\mu_0} \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t}$$

Step 3 — Replace the time derivatives with curls. From Ampère–Maxwell's law, $\partial_t \mathbf{E} = \frac{1}{\mu_0 \varepsilon_0} \nabla \times \mathbf{B}$; from Faraday's law, $\partial_t \mathbf{B} = -\nabla \times \mathbf{E}$:

$$\frac{\partial u}{\partial t} = \frac{1}{\mu_0} \mathbf{E} \cdot (\nabla \times \mathbf{B}) - \frac{1}{\mu_0} \mathbf{B} \cdot (\nabla \times \mathbf{E})$$

Step 4 — Recognize the divergence identity $\nabla \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{B})$:

$$\frac{\partial u}{\partial t} = -\nabla \cdot \left(\frac{\mathbf{E} \times \mathbf{B}}{\mu_0} \right)$$

Step 5 — Read off the continuity equation. This is $\partial_t u + \nabla \cdot \mathbf{S} = 0$ — local conservation of energy with no sources or sinks — where the flux is the **Poynting vector**:

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}$$

(units: W/m²)

Interpretation: $\mathbf{S}$ is the energy flux — the rate at which electromagnetic energy crosses a unit area, flowing in the direction of $\mathbf{E} \times \mathbf{B}$.

6. Intensity of a plane wave

For a monochromatic plane wave traveling along $\hat{\mathbf{k}}$, one has $\mathbf{B} = (\hat{\mathbf{k}} \times \mathbf{E})/c$, so

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B} = \frac{1}{\mu_0 c} \mathbf{E} \times \hat{\mathbf{k}}$$

and therefore $|\mathbf{S}| = E^2/(\mu_0 c)$, pointing along $\hat{\mathbf{k}}$. Time-averaging over a period for $\mathbf{E} = \mathbf{E}_0 \cos(\mathbf{k} \cdot \mathbf{x} - \omega t)$, using $\langle \cos^2 \rangle = \frac{1}{2}$, gives the **intensity**:

$$I = \langle S \rangle = \frac{E_0^2}{2\mu_0 c} = \frac{1}{2} \varepsilon_0 c E_0^2$$

As a bonus, the same algebra shows that $\mathbf{S}/c^2 = \varepsilon_0 \mathbf{E} \times \mathbf{B}$ is the electromagnetic **momentum density**: light carries momentum as well as energy.

7. Summary of results

Quantity	Expression
Wave equation (both fields)	$\nabla^2 \mathbf{F} = \frac{1}{c^2} \frac{\partial^2 \mathbf{F}}{\partial t^2}, \mathbf{F} \in \{\mathbf{E}, \mathbf{B}\}$
Speed of propagation	$c = 1/\sqrt{\mu_0 \varepsilon_0}$
Dispersion relation	$\omega = c \mathbf{k} $
Energy density	$u = \frac{1}{2} \varepsilon_0 E^2 + \frac{1}{2\mu_0} B^2$
Poynting vector	$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}$
Energy conservation	$\partial_t u + \nabla \cdot \mathbf{S} = 0$
Momentum density	$\mathbf{g} = \mathbf{S}/c^2 = \varepsilon_0 \mathbf{E} \times \mathbf{B}$
Intensity (monochromatic plane wave)	$I = E_0^2/(2\mu_0 c) = \frac{1}{2} \varepsilon_0 c E_0^2$